

A Mechanism of the Anodic Dissolution of Armco Iron and High-Strength Ferritic Cast Iron in an Oxalate Medium

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Abstract—The kinetics of the anodic dissolution of Armco iron and high-strength ferritic cast iron is investigated in an oxalic-acid medium at different pH values. It is shown that passivation occurs with the participation of secondary and electrochemical mechanisms and the formation of ferrous oxalate film. It is ascertained that hydro-oxalate ions participate in the electrode process with the reaction order equal to 0.5. The proposed mechanism of anodic dissolution of Armco iron and high-strength ferritic cast iron includes the stages of the formation of passivating film and Fe^{2+} ions. The latter, accumulating in a near-electrode layer, promote the passivation of metal by the mechanism of secondary crystallization.

It is commonly known that the passivation of metal in an acidic medium may be of secondary type, i.e., occur via the supersaturation of the near-electrode layer with metal salt and its surface precipitation [1]. The anodic dissolution of iron in oxalic-acid medium is accompanied by the formation of a film of difficultly soluble ferrous oxalate (II) [2, 3]. Films containing this salt are used as primers for paint coatings [4], and that is an additional reason for studying the salt passivity of iron in an oxalate medium. According to [2, 3], the formation of ferrous-oxalate protective layer includes a dissolution stage, although no detail is yet clear. Such a “space” mechanism of the salt film formation depends on the regularities of active dissolution leading to the saturation of the near-electrode layer with metal ions. Hence, it is reasonable to begin with a study of the metal behavior in a range of its active dissolution potentials. High-strength ferritic cast iron has a ferritic matrix, and its structure visually differs from that of Armco iron by the presence of spherical (rounded) graphite inclusions. The passivation of both the metals in an oxalic-acid medium may well be caused by the formation of a protective salt film, and investigating the mechanism of the film formation is the objective of this work.

EXPERIMENTAL

Figure 1 presents the microstructure of Armco-iron and cast iron samples we tested. Preparing the working surface, as well as measuring procedure itself, was described earlier in study [5]. Solutions of oxalic acid and potassium oxalate in different proportions, but with a constant ionic force equal to 0.48, were used as test media.

Potentiodynamically recorded (1 mV/s) voltammetric anodic curves of both the metals (Fig. 2) show that the metal passivation caused, as shown in [2], by the

formation of ferrous oxalate sets in at the electrode at a potential higher than E_{bp} and the corresponding current density j_{max} (table).

According to study [2], the passivation of iron is due to the supersaturation of the near-electrode electrolyte layer with Fe^{2+} ions followed by the crystallization of FeC_2O_4 . The concentration of Fe^{2+} ions is determined by the solubility product of ferrous oxalate and the concentration of oxalate ions: $[\text{Fe}^{2+}] = \frac{\text{PP}_{\text{FeC}_2\text{O}_4}}{[\text{C}_2\text{O}_4^{2-}]}$. Since the

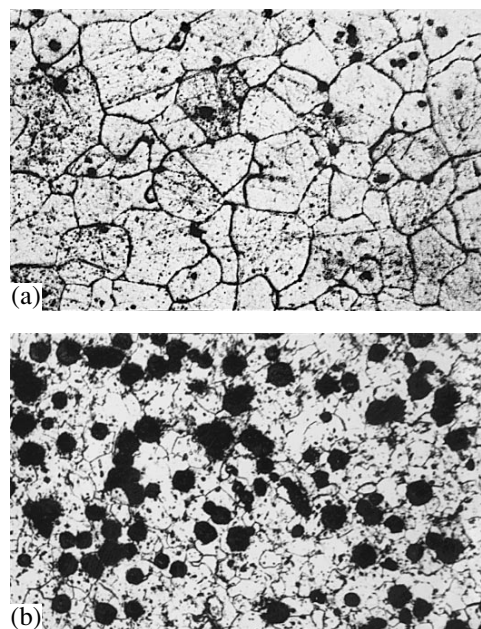


Fig. 1. Microstructure of (a) Armco iron and (b) high-strength ferritic cast iron. Pickling with 3% alcoholic HNO_3 solution for 8 s. Magnification 100.

The effect of pH and the concentration of anionic forms of oxalic acid on the maximum anodic current ($j_{\max}$) and the passivation potential (E_{bp}) of Armco iron and high-strength ferritic cast iron

pH	$\log[\text{H}_2\text{C}_2\text{O}_4]$	$\log[\text{HC}_2\text{O}_4^-]$	$\log[\text{C}_2\text{O}_4^{2-}]$	$j_{\max} \times 10^3, \text{ A/cm}^2$		$E_{\text{bp}}, \text{ V}$	
				armco iron	cast iron	armco iron	cast iron
1.25	-1.107	-1.087	-3.927	2.19	1.87	-0.153	-0.154
1.40	-1.191	-1.021	-3.711	—	1.52	—	-0.148
2.00	-1.677	-0.867	-2.957	1.38	1.18	-0.141	-0.173
3.50	-3.167	-0.897	-1.487	1.84	—	-0.220	—
3.72	-3.441	-0.951	-1.321	—	2.02	—	-0.230
4.68	-4.396	-1.485	-0.895	—	3.25	—	-0.261
4.90	-5.338	-1.668	-0.858	2.58	3.47	-0.308	-0.251
5.20	-5.908	-1.938	-0.828	2.61	4.70	-0.308	-0.218

concentration of C_2^{2-} ions increases, in our solutions, with the pH (table), the Fe^{2+} concentration necessary to start the crystallization of FeC_2O_4 should decrease. Hence, because the formation of Fe^{2+} ions is essentially the summary anodic reaction, the electric charge (Q_a) spent on the passivation of Armco and cast iron should decrease with an increase in pH.

We determined the Q_a values by numerically integrating the potentiodynamic curves in current density–time coordinates with the method of parabolas [6] in a range from 0 to $t_{\max}$ (i.e., to $j_{\max}$, Fig. 2b). The obtained dependence of Q_a on pH (Fig. 3) consists, however, in the increase rather than decrease in Q_a with an increase in pH. Hence, a direct electrochemical mechanism of the formation of FeC_2O_4 is probable. Insofar as ferrous

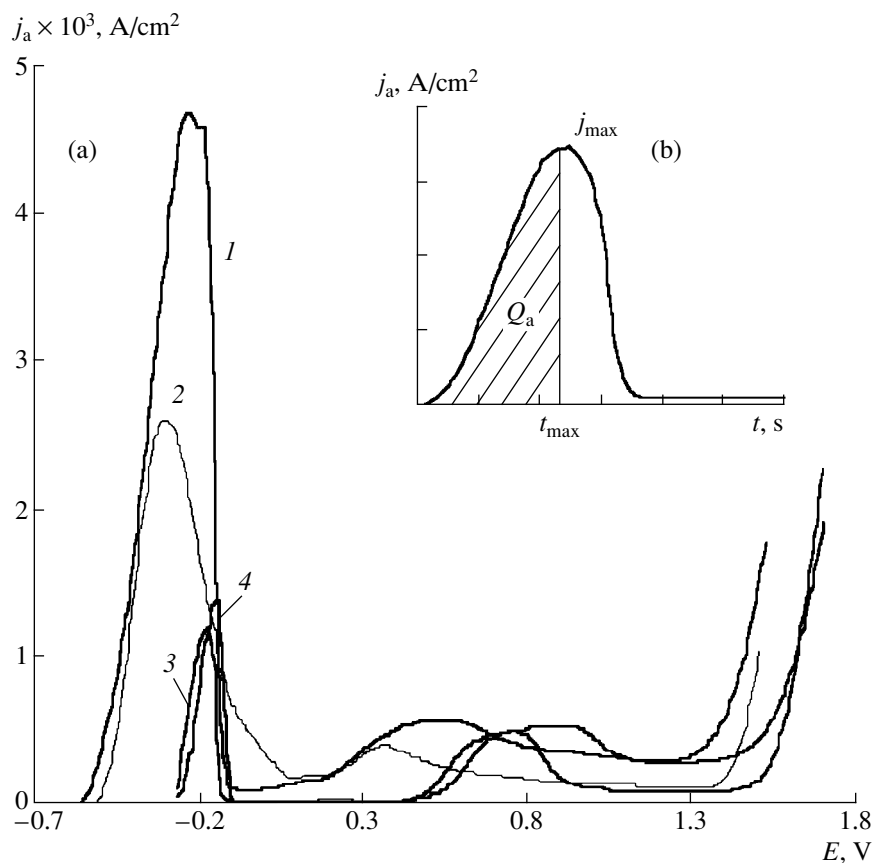
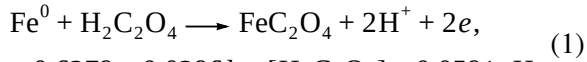
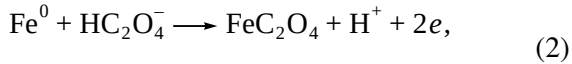


Fig. 2. (a) Voltammetric curves of (1, 3) Armco iron and (2, 4) cast iron at pH 2 and 5.2, respectively, and (b) calculating the electric charge by the numerical integration method.

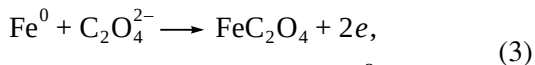
oxalate can electrochemically form on ferritic areas of metal, which are inherent in both the investigated materials (Fig. 1), we can assume one and the same mechanism of their anodic dissolution in oxalate medium. Oxalic acid can exist in solution in the form of undissociated molecules ($\text{H}_2\text{C}_2\text{O}_4$), hydro-oxalate (HC_2O_4^-), and oxalate ions ($\text{C}_2\text{O}_4^{2-}$), and each can interact with iron to form ferrous oxalate (II):



$$E = -0.6279 - 0.0296 \log[\text{H}_2\text{C}_2\text{O}_4] - 0.0591 \text{pH};$$



$$E = -0.6643 - 0.0296 \log[\text{HC}_2\text{O}_4^-] - 0.0296 \text{pH};$$



$$E = -0.7849 - 0.0296 \log[\text{C}_2\text{O}_4^{2-}].$$

Calculating empirical coefficients $\partial E/\partial \text{pH}$ and comparing them with the corresponding theoretical coefficients for equations (1)–(3) can help in judging which form of oxalic acid participates in the process.

Taking into account the generalized equation of the electrode potential

$$E = E^0 + A \log C_i + B \text{pH}, \quad (4)$$

where E^0 is the standard electrode potential; C_i is the concentration of the anionic form of acid (while A and B are coefficients), we proceeded from concentrations of a specified anionic form and the potential values E_{bp} corresponding to j_{max} (see table) to calculate, using the multiple regression analysis [7], the coefficients $B = \partial E/\partial \text{pH}$, which are given together with theoretical ones as follows:

Form	$\text{H}_2\text{C}_2\text{O}_4$	HC_2O_4^-	$\text{C}_2\text{O}_4^{2-}$
C_{calc}	-0.094	-0.036	+0.020
C_{theor}	-0.059	-0.030	0

The value $B = -0.036$ is the nearest to the theoretical one, corresponding to the process described by equation (2) with the participation of hydro-oxalate ions (-0.030). The coefficients for nondissociated molecules and oxalate ions sharply differ from those for processes described by Eqs. (1) and (3), which allows one to assume that HC_2O_4^- ions participate in the electrochemical formation of a passivating film of ferrous oxalate (II).

RESULTS AND DISCUSSION

At fixed anodic polarization values, the dependence of the logarithm of anodic current of both the metals on the pH of solution (Fig. 4) consists of three rectilinear

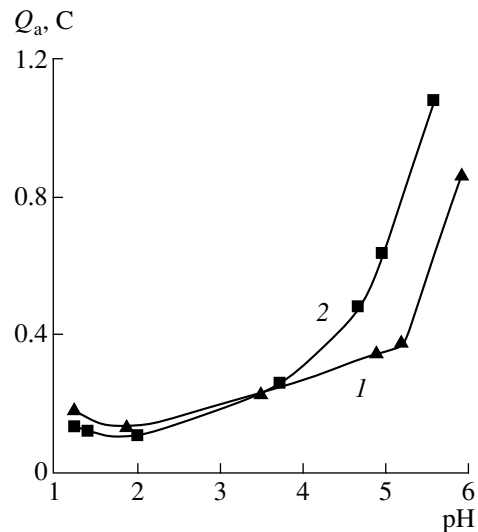


Fig. 3. Dependence of the electric charge Q_a on pH for (1) Armco iron and (2) cast iron.

segments with different slopes, which proves that not all anionic forms of oxalic acid participate equally in the anodic reaction. The identity of the dependences $\log j_a - \text{pH}$ for Armco and cast iron substantiates a hypothesis on the identical mechanism of their anodic dissolution. The kinks (Fig. 4) at pH 1.23 and 4.09 correspond to equality of molar fractions of either $\text{H}_2\text{C}_2\text{O}_4$ and HC_2O_4^- or HC_2O_4^- and $\text{C}_2\text{O}_4^{2-}$, respectively (Fig. 5). However, insofar as different concentrations of the anionic forms of the acid correspond to different experimental pH values (Fig. 5), the slope of none of the three segments of the $\log j_a - \text{pH}$ dependence (Fig. 4) can equal the reaction order in H^+ ions. An analysis of the experimental data shows that, with an increase in pH from 2.00 to 3.72, i.e., 1.9 times, the concentration of HC_2O_4^- ions changes by only a factor of 1.2, while the anodic current values at these points are close to each other (Fig. 4). This makes us to conclude that it is hydro-oxalate ions that participate in the reaction, while the participation of hydrogen ions is not manifest. In a pH range from 4.0 to 5.7, the concentrations of HC_2O_4^- and $\text{C}_2\text{O}_4^{2-}$ ions change fourfold and less than by 20%, respectively, which again relates the change in the anodic current in this range (Fig. 4) to the participation of the hydro-oxalate ions in the process. The oxalate ions do not participate in the process, since at pH 2.00 and 3.72 the anodic current is the same (Fig. 4), whereas the concentration of oxalate ions changes substantially (Fig. 5).

Thus, the analysis of the $\log j_a - \text{pH}$ dependence (Fig. 4) leads to a conclusion that only hydro-oxalate and, probably, H^+ ions participate in the anodic dissolution of Armco iron and high-strength ferritic cast iron

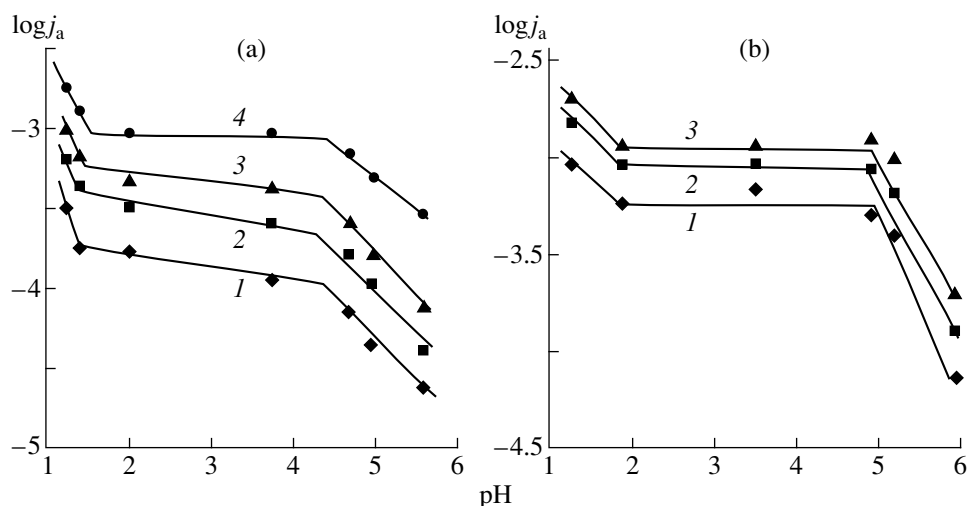


Fig. 4. Dependence of the logarithm of anodic current for (a) Armco iron and (b) cast iron on pH at an anodic polarization (mV) of (1) 40, (2) 60, (3) 80, and (4) 100.

in an oxalic-acid medium. In such a case, the basic kinetic equation of the anodic dissolution process will be as follows:

$$j_a = K[\text{HC}_2\text{O}_4^-]^{n_1} \cdot [\text{H}^+]^{n_2} \cdot \exp\left(\frac{\alpha F}{RT} \Delta E\right), \quad (5)$$

where K is the reaction rate constant; n_1 and n_2 are the reaction orders in HC_2O_4^- and H^+ ions, respectively; ΔE is the shift from the free corrosion potential; and α is the transfer coefficient. The slopes of rectilinear segments of the anodic ΔE – $\log j$ curve (b_a) equal 0.045 V for both Armco iron and cast iron. From the expression for the single-electron stage ($z = 1$), the transfer coefficient α is determined by the relation $\alpha = \frac{2.3RT}{b_a F}$ [8],

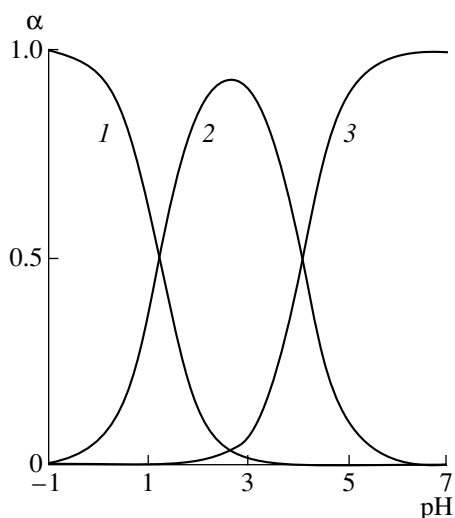


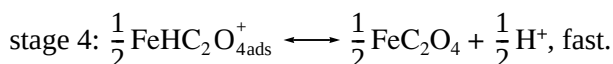
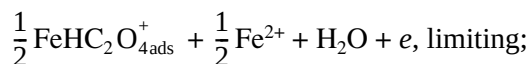
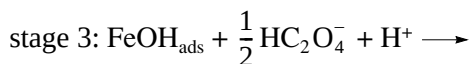
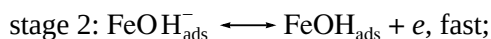
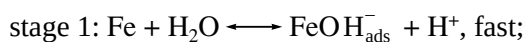
Fig. 5. Dependence of molar fractions of (1) nondissociated molecules of oxalic acid, (2) hydro-oxalate, and (3) oxalate ions on pH.

and at $b_a = 0.045$ we obtain $\alpha = 1.31$. This is undoubtedly an apparent value, which is characteristic of none of the stages of the process, and can be observed when the process includes several closely fast chemical or electrochemical stages [8].

The reaction orders in hydro-oxalate and hydrogen ions, n_1 and n_2 can be calculated with an iterative method [9] which is implemented as the SIMPLEX method in the STATISTICA computer program. In our case, the initial data are the concentrations of hydro-oxalate ions and protons, while the response function is derived from the anodic current for the Armco iron and ferritic cast iron (see table). The calculation results show that $n_1 = 0.5$, and $n_2 = 0.05$ for both the metals, whence it follows that the concentration of H^+ ions does not figure in the kinetic equation of the total anodic reaction. In view of the calculated values of n_1 , n_2 , and α , the basic equation (5) becomes

$$j_a = K \cdot [\text{HC}_2\text{O}_4^-]^{0.5} \cdot \exp\left(\frac{1.31F}{RT} \Delta E\right). \quad (6)$$

Thus, Eq. (6) determines the slow electrochemical stage of the following proposed mechanism of the anodic dissolution process:



In the mechanism proposed, the chemical stage (1) consists in the formation of adsorbed charged particles $\text{FeOH}_{\text{ads}}^-$ [8] with a concentration $[\text{FeOH}_{\text{ads}}^-] = K_1/[\text{H}^+]$, where K_1 is the process rate constant.

The second, fast stage (2) is electrochemical and consists in the detachment of one electron from a $\text{FeOH}_{\text{ads}}^-$ particle.

The third, limiting stage (3) determines the rate of the whole process; the kinetic equation of the stage is as follows:

$$j_3 = K_3[\text{FeOH}_{\text{ads}}^-] \cdot [\text{HC}_2\text{O}_4]^{0.5} \cdot [\text{H}^+] \cdot \exp\left(\frac{\alpha_3 F}{Rt} \Delta E\right). \quad (7)$$

If substitute the $\text{FeOH}_{\text{ads}}^-$ concentration calculated from the stage (2) $\text{FeOH}_{\text{ads}}^- = K_2[\text{FeOH}_{\text{ads}}^-] \exp\left(\frac{F}{Rt} \Delta E\right)$, where K_2 is the rate constant of the second stage, and $\text{FeOH}_{\text{ads}}^-$ concentration from the stage (1) to Eq. (7), we obtain the following expression:

$$j_3 = K_3 K_2 \frac{K_1}{[\text{H}^+]} \cdot \exp\left(\frac{F}{RT} \Delta E\right) \cdot [\text{HC}_2\text{O}_4]^{0.5} \cdot [\text{H}^+] \times \exp\left(\frac{\alpha_3 F}{RT} \Delta E\right) \Rightarrow j_3 = K[\text{HC}_2\text{O}_4]^{0.5} \times \exp\left(\frac{(1 + \alpha)F}{RT} \Delta E\right), \quad (8)$$

where $K = K_3 K_2 K_1$, and $\alpha = \alpha_3$.

Assuming $\alpha \approx 0.5$, we obtain the "apparent" transfer coefficient $(1 + \alpha)$ equal to 1.5, which is close to the experimental value 1.31 determined from the slope of the rectilinear segment of the polarization curve. From Eq. (8), we obtain the anodic slope value $b_a = 0.040$ V; its experimental value is $b_a = 0.045$ V. Using the same equation (8), we obtain the reaction order 0.5 in hydro-oxalate ions, and zero in H^+ ions.

Now, the reaction order, transfer coefficient, and slope b_a , calculated by Eq. (8) corresponding to the process mechanism proposed coincide with the experimental values. This substantiates the correctness of the scheme proposed for the process of the anodic dissolution of Armco iron and high-strength ferritic cast iron in oxalic-acid media. According to the limiting stage (3), it is not the whole amount of current supplied to the

electrode that is consumed for the electrochemical formation of the passivating layer; a part of it is spent on the formation of Fe^{2+} ions, which is confirmed by the dependence of the Q_a value on pH (Fig. 3). The slow stage (3) combines the electrochemical and secondary mechanisms of passivation, because the passive layer and Fe^{2+} ions are the conjugate reaction products and, accumulating in the near-electrode layer, the ions facilitate the passivation of metal.

CONCLUSIONS

(1) A mechanism of the dissolution of Armco iron and high-strength ferritic cast iron comprising four stages, the formation of $\text{FeHC}_2\text{O}_4^+$ ion adsorbed on the iron surface being the limiting one, is proposed.

(2) It is assumed that the passive ferrous oxalate layer and Fe^{2+} ions are conjugate products formed at the same stage of the process so that electrochemical and secondary mechanisms merge in the scheme of the dissolution and passivation of Armco iron, as well as high-strength ferritic cast iron, in an oxalic-acid medium.

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